

Static Website Deployment: Azure Static Web Apps & GitHub Actions (CI/CD)

This guide documents the process I followed for the Azure Fundamentals task, detailing how I deployed a simple personal portfolio website from a private GitHub repository to Azure Static Web Apps (SWA). The entire CI/CD process is automated using the integrated GitHub Actions workflow.

Goal: To establish an automated CI/CD pipeline where any push to the main branch on GitHub automatically builds and deploys the latest version of my website to the live Azure URL.

Step 1: GitHub Repository Setup

To begin, I created a private GitHub repository named `private-portfolio` to host my project's source code. The project is a simple static website built with HTML, CSS, and JavaScript.

The key to this deployment is understanding my file structure:

My `index.html` file is located in the root of the repository.

All supporting assets (css, js, and images files) are organized inside an `/assets` folder, which is also in the root.

/ (Repository Root)

```
├── .github/
│   └── workflows/
│       └── azure-web-static.yml (This file was auto-generated by Azure in Step 2)
├── assets/
│   ├── css/
│   │   └── style.css
│   ├── js/
│   │   └── script.js
│   └── images/
│       └── ...
└── index.html
```

Step 2: Creating the Azure Static Web App Resource

In this step, I created the core Azure resource that would host my site and connected it directly to my GitHub repository.

2.1 Creating the Resource:

I navigated to the Azure Portal and searched for "Static Web Apps".
I selected my Subscription and Resource Group.
I gave my app a unique Name (onur-personal-portfolio).
I selected a Region close to me.(West Europe)
I chose the "Standart" plan for this task.

Create Static Web App ...

The screenshot shows the 'Create Static Web App' page in the Azure Portal. The 'Project Details' section is active, showing the 'Subscription' as 'Azure subscription 1' and the 'Resource Group' as 'onurEpam'. Below this, the 'Hosting region' section shows 'Static Web Apps distributes your app's static assets globally. Configure regional features in [Advanced](#)'. The 'Regions' section shows 'Global' selected. The 'Static Web App details' section shows the 'Name' as 'onur-personal-portfolio' with a green checkmark.

[2.1 Image]

2.2 Deployment Details & GitHub Connection:

In the "Deployment details" section on the same page, I selected "GitHub" as the source.
I clicked the "Sign in with GitHub" button to authorize Azure to access my repositories.
Organization: I selected my GitHub username.
Repository: I selected my private-portfolio repository.
Branch: I selected main.Finally, I clicked "Review + create" and then "Create".

The screenshot shows the 'Deployment details' section of the 'Create Static Web App' page. The 'Hosting plan' section shows 'Standard: For general purpose production apps' selected. The 'Deployment details' section shows 'Source' as 'GitHub'. The 'GitHub account' is 'WarLock54' with a 'Change account' button. A blue information banner states: 'If you can't find an organization or repository, you might need to enable additional permissions on GitHub. You must have write access to your chosen repository to deploy with GitHub Actions.' Below this, the 'Organization' is 'WarLock54', the 'Repository' is 'privatePortfolio', and the 'Branch' is 'main'.

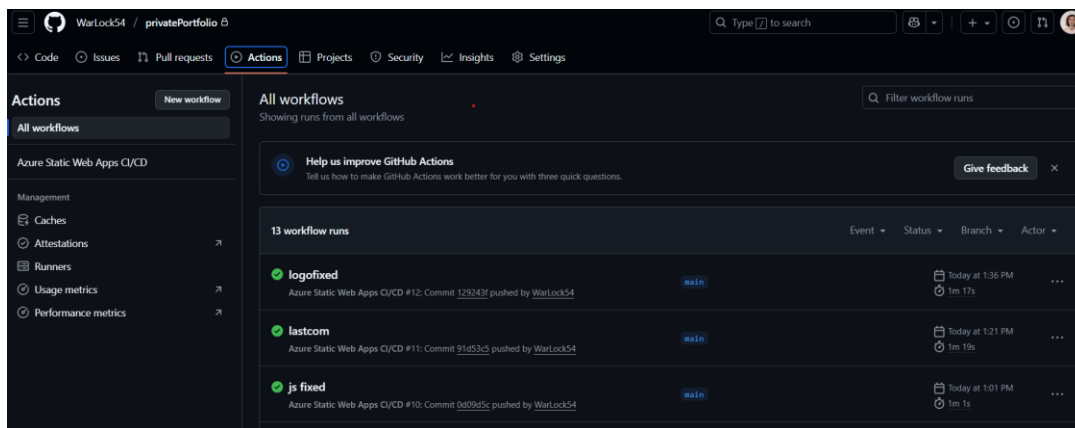
[2.2 Image]

Step 3: Configuring the GitHub Actions Workflow (.yml) for CI/CD

When I created the SWA source, Azure automatically saved a new workflow file (.yml file) to my repository under .github/workflows/. After that, it wasn't seeing my javascript file, so I used the document.addEventListener method to delay execution until the HTML loaded.

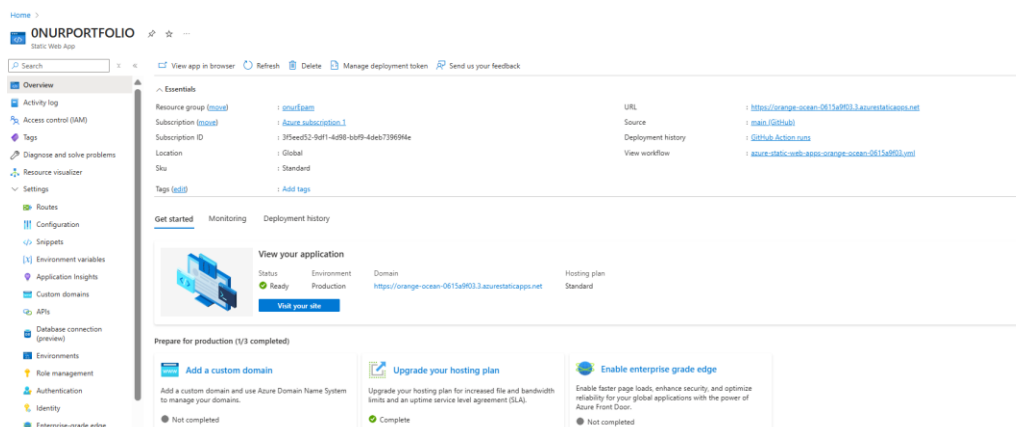
Step 5: Results and Verification (Task Complete)

I committed and pushed both the js and index.html file changes to my main branch. This push automatically triggered a new GitHub Actions run. I went to the "Actions" tab in my GitHub repository and watched as the workflow successfully completed (it showed a green checkmark).



[5.1 Image]

I returned to the "Overview" page for my Static Web App resource in the Azure Portal. Result: My personal portfolio website loaded perfectly. All CSS styles were applied, and all JavaScript interactions (like button clicks) were fully functional. The task was successfully completed, and the CI/CD pipeline is now active.



[5.2 Image]

Live Site URL: <https://orange-ocean-0615a9f03.3.azurestaticapps.net/>

